

Tasks 1: Use the ping command to test connectivity from your computer to [www.youtube.com](http://www.youtube.com) and note the average response time.

```
C:\Users\srikantm05>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4829:400::] with 32 bytes of data:
Reply from 2001:4860:4829:400::: time=71ms
Reply from 2001:4860:4829:400::: time=93ms
Reply from 2001:4860:4829:400::: time=122ms
Reply from 2001:4860:4829:400::: time=61ms

Ping statistics for 2001:4860:4829:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 61ms, Maximum = 122ms, Average = 86ms
```

Avg Response time: 86ms

Task 2: Run the traceroute (tracert on Windows) command to [www.flipkart.com](http://www.flipkart.com) and list all the intermediate hops shown in the output.<br><br><em><strong>Hint:</strong></em> Each hop represents a network device your data passes through to reach the destination.

```
C:\Users\srikantm05>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.55.244.187]
over a maximum of 30 hops:

  1   11 ms    3 ms    4 ms   10.66.213.140
  2  229 ms   242 ms  201 ms  192.168.17.10
  3   64 ms   104 ms   63 ms  192.168.16.49
  4  176 ms    78 ms    78 ms  192.168.19.23
  5   48 ms    80 ms    78 ms  192.168.19.49
  6   62 ms    76 ms    82 ms  192.168.252.34
  7   82 ms    79 ms    66 ms  125.20.209.146
  8   62 ms    74 ms    91 ms  125.20.209.145
  9   62 ms   114 ms   48 ms  116.119.36.180
 10  124 ms    77 ms    *     ae8.gpx-bom2.netarch.akamai.com [23.55.247.248]
 11   84 ms    71 ms    84 ms  a23-55-244-187.deploy.static.akamaitechnologies.com [23.55.244.187]

Trace complete.
```

Task 3: Use the nslookup tool to find the IP address of [www.zomato.com](http://www.zomato.com) and record the result.

```
C:\Users\srikantm05>nslookup www.zomato.com
Server:  reliance.reliance
Address: 2405:201:200f:c1cf::c0a8:1d01

Non-authoritative answer:
Name:    e70929.dsca.akamaiedge.net
Addresses: 2600:140f:7200:49::17d4:65
           2600:140f:7200:49::17d4:73
           49.44.140.187
           49.44.140.145
Aliases: www.zomato.com
          zomato.edgekey.net
```

**What does 'nslookup' do:** The computer asks to look up the IP address for [www.zomato.com](http://www.zomato.com).

**What happened, step by step:**

1. The computer asked ISP's DNS server (reliance.reliance) — "what's the address for [www.zomato.com](http://www.zomato.com)?"
2. Instead of giving one straight answer, the DNS server said: "[www.zomato.com](http://www.zomato.com) is actually just a nickname for [zomato.edgekey.net](http://zomato.edgekey.net), which is itself a nickname for [e70929.dsca.akamaiedge.net](http://e70929.dsca.akamaiedge.net)
3. That final name is the one that actually has real addresses attached to it.

**Why the nicknames (aliases)?** Zomato doesn't host its website on one single server. It uses a service called **Akamai**, which runs thousands of servers all over the world. When you type [zomato.com](http://zomato.com), Akamai figures out which of its servers is closest/fastest for you and hands you that address. That's why the name changes hands a few times before landing on a real IP — it's Zomato outsourcing "which server should this person connect to" to Akamai.

**There are 4 addresses instead of 1:**

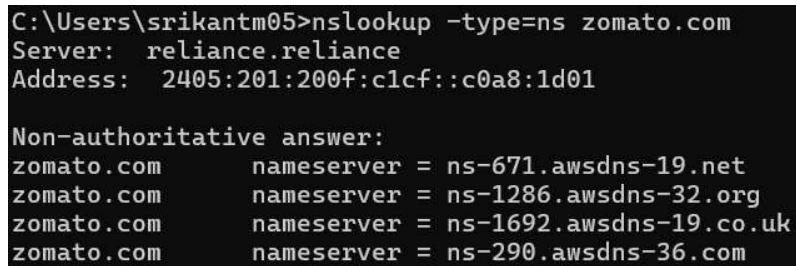
- Two of them (49.44.140.187 and 49.44.140.145) are old-style addresses (IPv4) — the common kind.
- Two of them (2600:140f:...) are new-style addresses (IPv6)

- Having more than one of each type is just backup — if one server is busy or down, your device can try the next one.

**"Non-authoritative answer"** just means: your ISP DNS server didn't personally go ask Zomato/Akamai this very second — it's giving you an answer it saved from a recent lookup.

### Finding the authoritative name servers

Command used: `nslookup -type=ns zomato.com`



```
C:\Users\srikantm05>nslookup -type=ns zomato.com
Server:  reliance.reliance
Address:  2405:201:200f:c1cf::c0a8:1d01

Non-authoritative answer:
zomato.com       nameserver = ns-671.awsdns-19.net
zomato.com       nameserver = ns-1286.awsdns-32.org
zomato.com       nameserver = ns-1692.awsdns-19.co.uk
zomato.com       nameserver = ns-290.awsdns-36.com
```

Figure 3: Authoritative name servers for zomato.com

Result: zomato.com is delegated to four Amazon Route 53 name servers, spread across different top-level domains (.net, .org, .co.uk, .com):

- ns-671.awsdns-19.net
- ns-1286.awsdns-32.org
- ns-1692.awsdns-19.co.uk
- ns-290.awsdns-36.com

Meaning: A top-level domain (TLD) is the last part of a domain name, such as .com, .org, or .in. Spreading name servers across four different TLDs is a redundancy measure. If one entire TLD's registry had a problem, the other three name servers on different TLDs would still be reachable, so the domain would keep working.

**Task 4:** Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

```
C:\Users\srikantm05>ping 192.168.29.5

Pinging 192.168.29.5 with 32 bytes of data:
Reply from 192.168.29.218: Destination host unreachable.
Reply from 192.168.29.218: Destination host unreachable.
Reply from 192.168.29.218: Destination host unreachable.
Reply from 192.168.29.218: Destination host unreachable.

Ping statistics for 192.168.29.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

When we ping a device, the computer first checks whether it knows the target's MAC address (hardware address) on the local network. If it doesn't, it asks the router. If the router also has no record of that device — because it's turned off, disconnected from Wi-Fi, or simply doesn't exist on that network — the router sends back a message saying "destination host unreachable" instead of silently ignoring the request. So the reply you see isn't from the device you're trying to reach; it's your router telling you it gave up looking for it.

Lost = 0 (0% loss) doesn't mean the ping succeeded. It means all 4 packets got *some* response — just not from the target device itself. The router answered every single request with an error message, so nothing was technically "lost" in transit; it simply came back as a rejection rather than a success.

Task 5: Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Constraint: Give a specific example for each tool based on your own test results above.

| Tool                 | Protocol it uses                         | OSI layer             | TCP/IP model layer |
|----------------------|------------------------------------------|-----------------------|--------------------|
| ping                 | ICMP (Internet Control Message Protocol) | Layer 3 – Network     | Internet layer     |
| nslookup             | DNS (Domain Name System)                 | Layer 7 – Application | Application layer  |
| tracert / traceroute | ICMP (Windows) or UDP + ICMP (Linux/Mac) | Layer 3 – Network     | Internet layer     |

### ping — "is it alive, and how fast?"

Sends small test packets to a target and waits for a reply:

- Whether the device responds at all
- Round-trip time (latency)
- Packet loss

### nslookup — "what's the address behind this name?"

Doesn't test connectivity at all — it only asks a DNS server to translate a domain name into an IP address (or vice versa). When used on `www.zomato.com` and it was found it resolves through Akamai's CDN. You'd use nslookup when the question is about *naming*, not *reachability* — e.g. "which server does this domain actually point to right now?"

#### (a) website not loading

- Website not loading → **start with nslookup, then ping** (DNS problems are a very common cause of "site won't load" that ping alone would miss)
- **ping** that IP to check if the server reachable
- if ping fails or times out use trace route to find out where does the path break.

**(b) Slow connection → traceroute** . If a connection is up but sluggish, ping provides the overall round-trip time to the final destination. Traceroute breaks that same round trip into hop-by-hop timings, so you can see exactly which router along the path is adding delay — your own router, your ISP's network, or somewhere further out. That's the right diagnostic tool for "it's slow, but where is it slow."

**(c) Suspecting DNS issues → nslookup** . This is nslookup's entire purpose: checking whether a domain name resolves to the IP address you expect. If a site "won't load" but pings fine when you use its IP directly, that's a strong sign the problem is DNS, not connectivity — and nslookup is exactly how to confirm it.